

(classnotes) 3.4 Function Composition

Notations you may see:

$$(f \circ g)(x) = f(g(x))$$

$$(f + g)(x) = f(x) + g(x)$$

$$(f - g)(x) = f(x) - g(x)$$

$$(fg)(x) = f(x)g(x)$$

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \quad \text{where } g(x) \neq 0$$

Review

For the following transformed functions, identify the base functions and the "mini function"

$$m(x) = \frac{1}{x+2}$$

$$n(x) = \sqrt{2x}$$

$$p(x) = (x - 4)^3$$

$$q(x) = \frac{1}{\sqrt{x}}$$

Practice using composition notation

Given $f(x) = x^2 - 9x$, and $g(x) = x + 9$

Calculate $f(g(x))$

Calculate $g(f(x))$

Calculate $g(g(x))$

Calculate $f(f(x))$

Given $f(x) = x^2 - 7x$, and $g(x) = x - 6$

Calculate $f(g(x))$

Calculate $g(f(x))$

Calculate $g(g(x))$

Calculate $f(f(x))$

Practice review: multiplying binomials

$$\begin{aligned}(x + 9)^2 &= (x + 9)(x + 9) \\ &= x^2 + 9x + 9x + 81 \\ &= x^2 + 18x + 81\end{aligned}$$

$$(x - 6)^2$$

$$(2x + 1)(x - 3)$$

$$(4x + 3)^2$$

$$(x - a)^2$$

$$(x^2 - 2x)(x + 1)$$

Using tables

Use the table below showing values of $f(x)$ and $g(x)$ to find each of the following. Remember when working with composition functions we always evaluate the inner function first.

x	-3	-2	-1	0	1	2	3
$f(x)$	11	9	7	5	3	1	-1
$g(x)$	-8	-3	0	1	0	-3	-8

- Ⓐ $f(1) =$
- Ⓑ $g(f(1)) =$
- Ⓒ $g(0) =$
- Ⓓ $f(g(0)) =$
- Ⓔ $f(g(2)) =$
- Ⓕ $f(f(3)) =$